

The empirical recurrence for OEIS A234792: recovering a conjecture that is not written down

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7 September 2026

Abstract

OEIS A234792 records “Empirical recurrence of order 66”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 76$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 66, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 66 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A234792 is “Number of $(n+1) \times (4+1)$ 0..3 arrays with each 2×2 subblock having the number of clockwise edge increases less than or equal to the number of counterclockwise edge increases.”. It has offset 1 and begins

371976, 102517088, 25695321008, 6215062828960, 1482624037868512, 351765183794721104, 832798148382242255

The entry states:

Empirical recurrence of order 66 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 08:27:17, revision 8) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1025$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 76$ of the 1025, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 76$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 2050$ exact terms of the model returns order 66 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{66} c_i a(n-i),$$

c_1	571	c_{18}	−1187240378507290382757657	c_{35}	1574888130700663523428076
c_2	−127728	c_{19}	3297004009492857114461236	c_{36}	−2304792107578454753594980
c_3	15627599	c_{20}	−1522769366239488005164585	c_{37}	3304290315990385418851287
c_4	−1209154423	c_{21}	−52716104074106033451797116	c_{38}	−4426130113544646822829521
c_5	64232622171	c_{22}	401348942292875765983192139	c_{39}	5204539398042604345925066
c_6	−2470695276575	c_{23}	−1878749208056812818619874388	c_{40}	−5130210603577527835764596
c_7	71428128199002	c_{24}	6568939471296368894517046696	c_{41}	41344865598795792522041466
c_8	−1595361434622326	c_{25}	−18033950241719384434435791779	c_{42}	−2689966072415422394786192
c_9	28114421414468609	c_{26}	38744288730097142110069182214	c_{43}	1381910217477881417883494
c_{10}	−397457759265378458	c_{27}	−59687064749361732365891690822	c_{44}	−503016398269001131660248
c_{11}	4569798231038056022	c_{28}	36332810486792655977584681421	c_{45}	503719268671934372383126
c_{12}	−43262971064959651304	c_{29}	144711753715846175156677550031	c_{46}	1008391964815237140620432
c_{13}	341655565906821093888	c_{30}	−680086211511394057857542588642	c_{47}	−954682134979581553169830
c_{14}	−2287024438997840604056	c_{31}	1811790961935555258870478364293	c_{48}	472857026448054196736724
c_{15}	13245519143028531712322	c_{32}	−3751321061693180171599145531173	c_{49}	−114137297864014152543804
c_{16}	−67769400613749640039795	c_{33}	6627057895804526054016060022639	c_{50}	−32619732034904874604882
c_{17}	307717011102816539754961	c_{34}	−10528743390394947253677952291479	c_{51}	543359690896071985556760

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 66, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $66 + 4$ terms removed returns order 66 as well, so $\deg q_{\min} = 66 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 10 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $66 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 66 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A234792.
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